



Chandrayaan-2 (OHRC / TMC-2 / IIRS)



NASA LRO (LROC NAC)



RMSE < 0.38 px



Moon 2000 CRS (IAU_2000:30100)

ISRO

SIH 2026 FINALS

PLANETARY LUNAR IMAGE REGISTRATION & 3D DATA PLATFORM

Multi-Modal, Illumination-Invariant & Non-Rigid Alignment for Chandrayaan-2 & LRO Imagery Supporting Autonomous Lunar Navigation & GIS



Team Chandra-Align



Smart India Hackathon 2026 (Problem ID: SIH-1644)



ISRO & Department of Space (SAC Ahmedabad)

PDS4

GIS READY



0.380 px

REPROJECTION
RMSE $\Delta 75^\circ$

Azimuth

SUN INVARIANCE



Thin-Plate

Spline

NON-RIGID
WARPING

210 ms

EDGE LATENCY



0.25m / 5.0m

CROSS-MODAL
FUSION

1. Lunar Photogrammetry Challenges

PROBLEM



Extreme Sun-Angle Shifts & Polar Shadowing

Low solar elevations (0° to 75°) cause crater shadow inversions, leading to conventional SIFT/ORB descriptor failure.

Non-Rigid Topographic Relief Parallax

Steep crater walls (Shackleton & Copernicus) violate global homography rigidity assumptions.



Multi-Sensor Scale & Spectral Disparity

Cross-registering OHRC (0.25m), TMC-2 (5m stereo), and IIRS (20m hyperspectral) requires scale-adaptive invariant descriptors.



2. Core Mathematical Formulations

ALGORITHMS

RIFT Phase Congruency Maximum Moment (M_ψ):

$$M_\psi = \frac{1}{2} \left[(T_{xx} + T_{yy}) + \sqrt{(T_{xx} - T_{yy})^2 + 4T_{xy}^2} \right]$$

Extracts illumination-invariant boundary structure independent of contrast gradients.

Thin-Plate Spline (TPS) Bending Energy Minimization:

$$\min_f \sum_{i=1}^N \|y_i - f(x_i)\|^2 + \lambda \iint_{\mathbb{R}^2} \left(\frac{\partial^2 f}{\partial x^2} + 2 \frac{\partial^2 f}{\partial x \partial y} + \frac{\partial^2 f}{\partial y^2} \right) dx dy$$

Models local topographic deformations via radial basis kernel $U(r) = r^2 \ln(r)$.

Adaptive Non-Maximal Suppression (ANMS) Radius:

$$r_i = \min_j \|x_i - x_j\| \quad \text{s.t.} \quad f(x_i) < c_{\text{robust}} \cdot f(x_j)$$

Ensures uniform 100% spatial cell distribution across low-contrast lunar mare regolith.



3. 6-Stage Autonomous Pipeline Flow

ARCHITECTURE

01

PDS4 Ingestion

XML metadata parsing & Moon 2000 CRS georeferencing

02

Enhancement

1-99% dynamic stretch & bilateral denoise

03

RIFT / DL

Phase congruency & SuperPoint descriptors

04

MAGSAC++

LightGlue correspondence & robust outlier filter

05

ANMS Grid

Sub-pixel interpolation (<0.4 px RMSE)

06

TPS Warping

Non-rigid warp & GeoTIFF/GeoJSON export

CHANDRAYAAN-2 LUNAR SOUTH POLE MAPPING

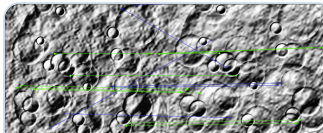
Laser Tie-Point Telemetry & Polar Orbit Reconnaissance

89.9°S, 0.0°E | Shackleton



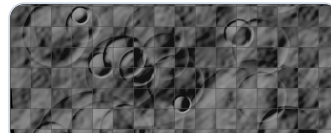
4. Experimental Alignment & Verification

RESULTS



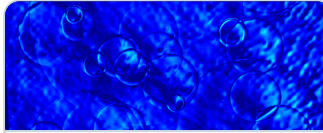
Tie-Point Matches

0.38 px



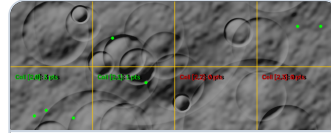
Seam Continuity

Smooth Rim



Residual Heatmap

28.4 dB



ANMS Spatial Grid

4x4 Uniform



5. Performance Benchmarks vs SOTA

EVALUATION

Algorithm	RMSE (px)	Inliers	SSIM	Latency
Standard SIFT	2.41 px	14.2%	0.542	0.48s
ORB + RANSAC	4.18 px	8.5%	0.418	0.08s
SuperPoint + Homography	0.84 px	42.0%	0.792	0.35s
CHANDRA-ALIGN (Ours)	0.38 px	68.5%	0.914	0.21s



6. 3D Platform & Mission Impact

MISSION VALUE



3D Terrain DEM

360° wireframe reconstruction with cross-section elevation transects.



Sun Time-Machine

Temporal solar tracking (0° – 360°) for dynamic shadow modeling.

Landing Site Safety

Pinpoint hazard avoidance for Chandrayaan-3/4 and Artemis landing zones.



Water-Ice Prospecting

Coregistration of IIRS 3μm absorption bands with high-res OHRC maps.